

Retail Sales & Discount Profitability Analysis

Furniture earns a 2.61% margin.
East Tables are the first priority for review.

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Furniture sales are substantial, but profit is thin

Business context and the decision for merchandising and finance

754,747.76

Furniture sales

19,730.00

Furniture profit

The decision

Which products and regions need a pricing and discount review before the next promotion?

Portfolio case based on supplied retail sales records.

Sales, profit and margin identify where discount policies need review

Scope: all records, followed by Furniture, Tables and Bookcases

Business question

Which product categories generate strong sales but weak profit, and where should discount policies be reviewed?

Sales = SUM(sales)

Profit = SUM(profit)

Margin = SUM(profit) / SUM(sales) × 100

Sales lines = COUNT(*)

Unique orders = COUNT(DISTINCT order_id)

Compare category, subcategory, discount group and region. Margins use aggregate totals.

10,194 sales lines represent 5,111 unique orders

Data preparation preserves source precision and all records

10,194

sales lines retained

0

duplicate Row IDs
or missing source values

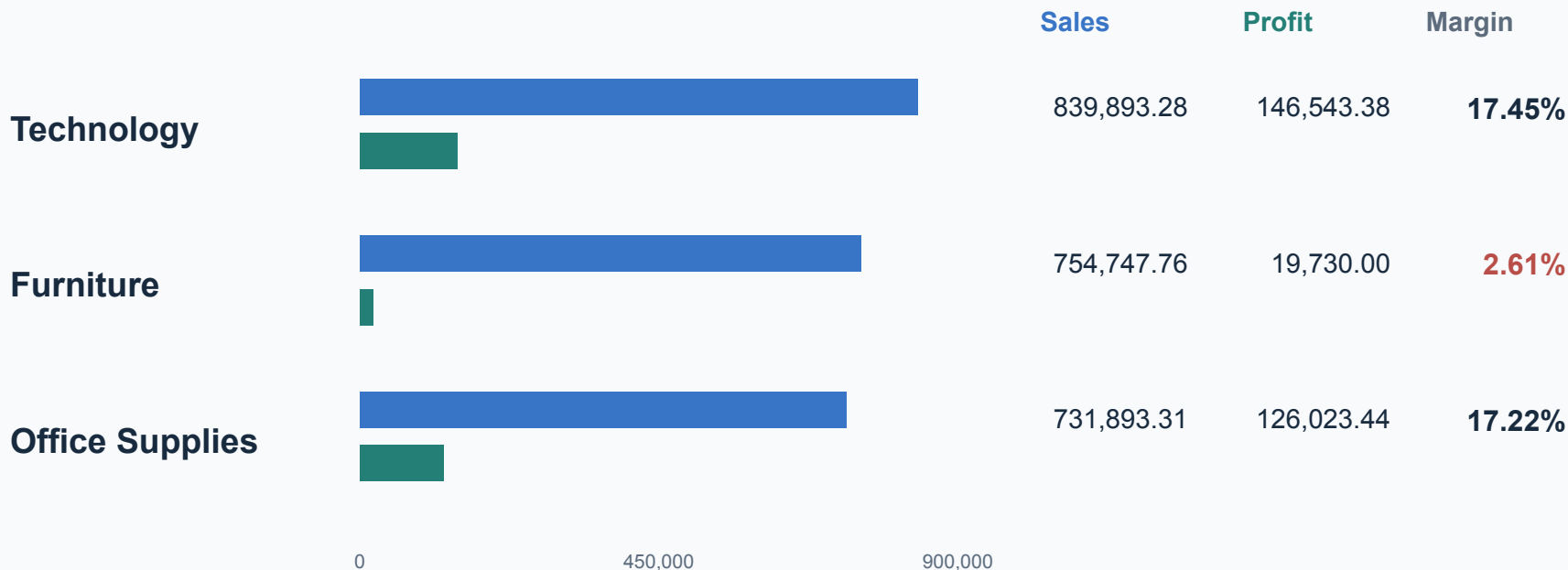
PostgreSQL workflow

Parse Order Date and Ship Date.
Cast Sales, Profit and Discount to numeric.
Create public.superstore_clean.
Aggregate using the supplied SQL groups.

Source period: 3 Jan 2023–30 Dec 2026.
Includes future-dated records relative to Sep 2026.
Currency is unspecified. No outlier rows removed.

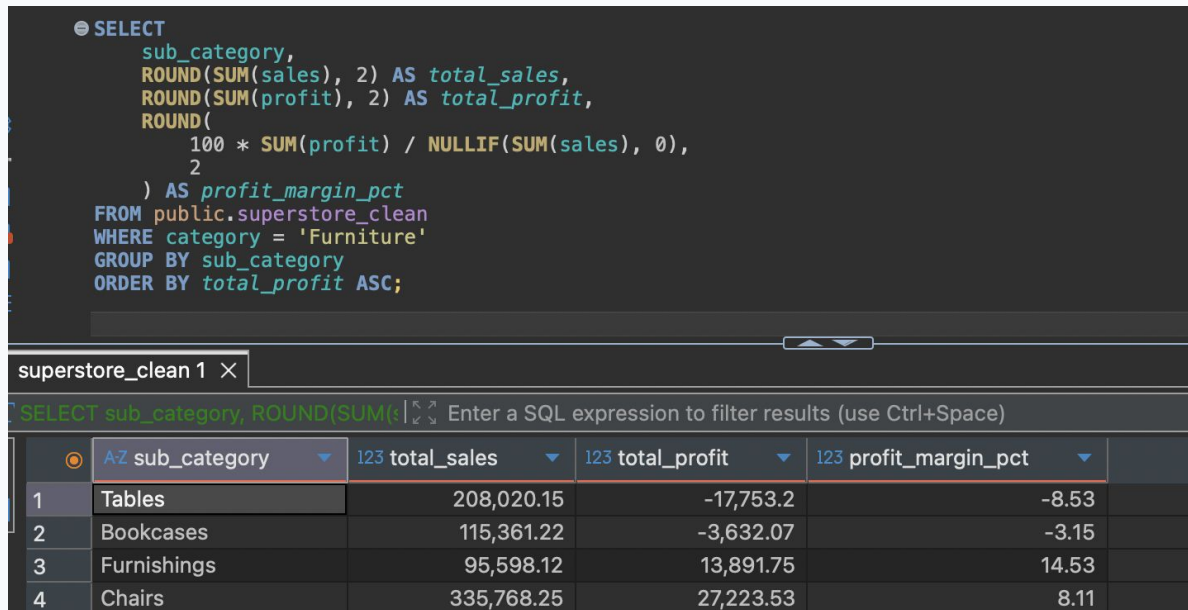
Furniture has 32.44% of sales, but only 6.75% of profit

Category comparison: sales and profit share the same scale

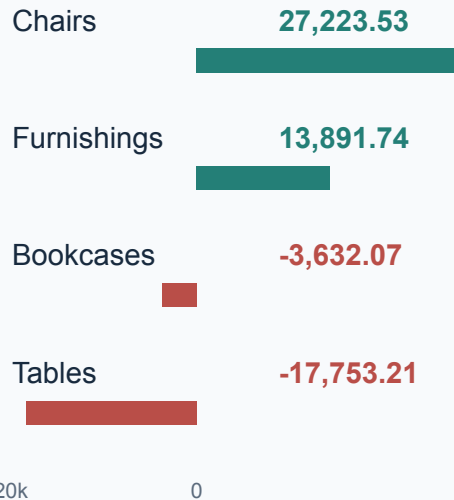


Tables and Bookcases reduce Furniture profit by 21,385.28

SQL groups Furniture sales lines by subcategory and calculates aggregate margins.



Profit from CSV

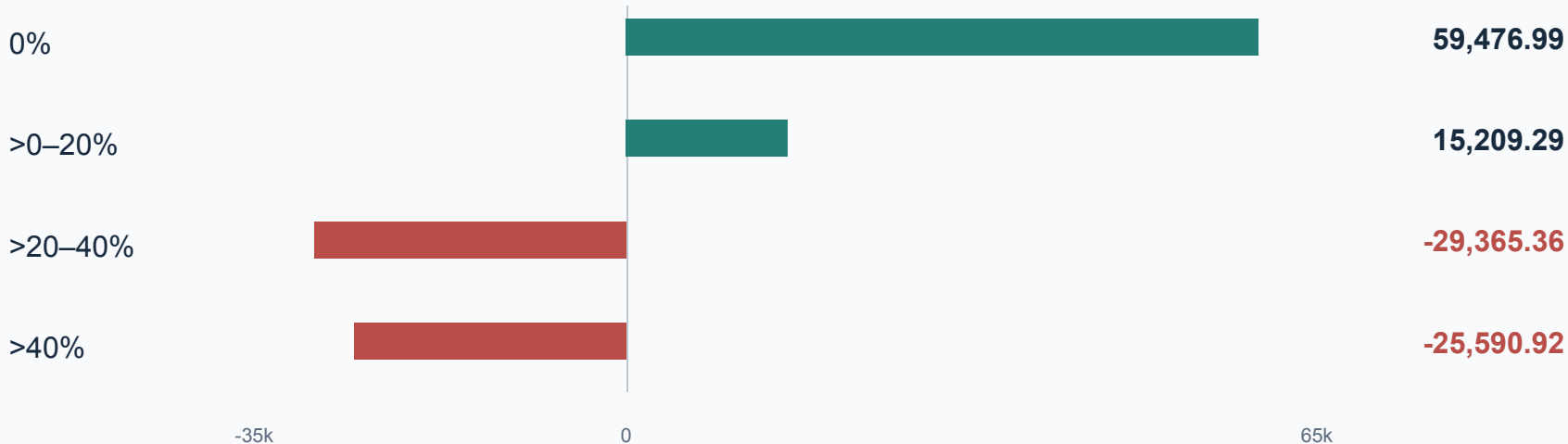


Profit, dataset units

Furniture discounts above 20% coincide with 54,956.29 in losses

Observed groups differ in their product mix and sales-line counts

Profit, dataset units



Sales lines by group: 880 / 766 / 326 / 229.

The >20% groups contain 555 lines. This comparison does not establish causation.

Tables show a loss even in the >0–20% discount group

```
SELECT
  sub_category,
  CASE
    WHEN discount_rate = 0 THEN '0%'
    WHEN discount_rate <= 0.20 THEN '>0–20%'
    WHEN discount_rate <= 0.40 THEN '>20–40%'
    ELSE '>40%'
  END AS discount_group,
  COUNT(*) AS sales_lines,
  ROUND(SUM(sales), 2) AS total_sales,
  ROUND(SUM(profit), 2) AS total_profit
FROM public.superstore_clean
WHERE sub_category IN ('Tables', 'Bookcases')
GROUP BY sub_category, discount_group
ORDER BY sub_category, MIN(discount_rate);
```

superstore_clean 1 X

SELECT sub_category, CASE WHEN d | Enter a SQL expression to filter results (use Ctrl+Space)

	AZ sub_category	AZ discount_group	123 sales_lines	123 total_sales	123 total_profit
1	Bookcases	0%	60	31,935.98	6,075.71
2	Bookcases	>0–20%	100	54,736.65	1,587.58
3	Bookcases	>20–40%	37	18,776.16	-2,947.01
4	Bookcases	>40%	35	9,912.43	-8,348.35
5	Tables	0%	76	71,706.23	13,334.93
6	Tables	>0–20%	72	45,636.57	-298.4
7	Tables	>20–40%	131	71,517.37	-19,681.23
8	Tables	>40%	47	19,159.99	-11,108.5

Tables: -298.40

Profit in the >0–20% group.
A 20% cap alone does not ensure profitability.

Both subcategories

show losses in each group above 20% discount.

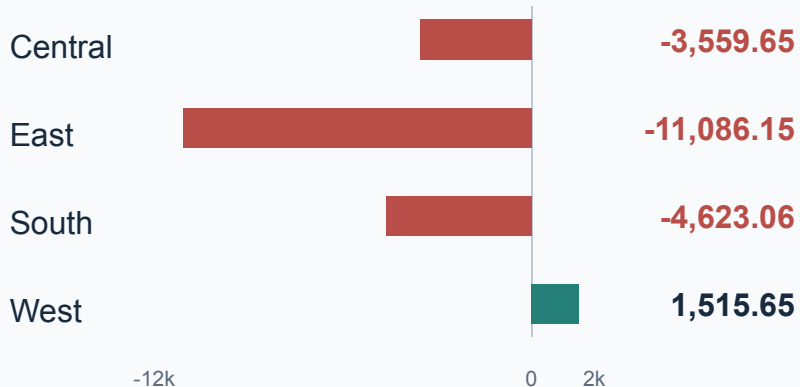
This pattern is descriptive.

East Tables show the largest loss among the regional combinations

Profit by region, using the same scale in both charts

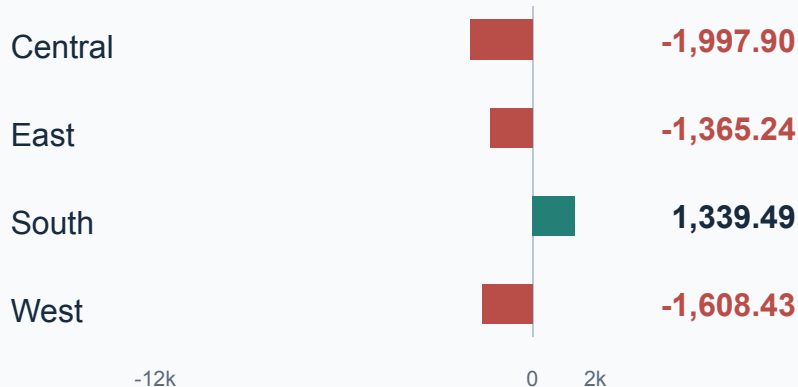
Tables

Profit, dataset units



Bookcases

Profit, dataset units



Tables earn positive profit only in West.

Bookcases earn positive profit only in South.

82 of 85 East Tables lines carry >20–40% discounts

```
SELECT
  CASE
    WHEN discount_rate = 0 THEN '0%'
    WHEN discount_rate <= 0.20 THEN '>0–20%'
    WHEN discount_rate <= 0.40 THEN '>20–40%'
    ELSE '>40%'
  END AS discount_group,
  COUNT(*) AS sales_lines,
  ROUND(SUM(sales), 2) AS total_sales,
  ROUND(SUM(profit), 2) AS total_profit
FROM public.superstore_clean
WHERE sub_category = 'Tables'
  AND region = 'East'
GROUP BY discount_group
ORDER BY MIN(discount_rate);
```

Results 1 ×

SELECT CASE WHEN discount_rate = | Enter a SQL expression to filter results (use Ctrl+Space)

	AZ discount_group	123 sales_lines	123 total_sales	123 total_profit
1	0%	3	65.51	30.74
2	>20–40%	82	39,860.62	-11,116.89

96.47%

of East Tables lines
have >20–40% discounts.

–11,116.89

profit in that group

Only 3 no-discount lines.
Too few for a reliable
comparison benchmark.

East Tables pricing needs review before the next promotion

Proposed actions and validation checks, with no assumed profit uplift

Commercial actions

Merchandising + Finance

Audit East Tables list prices, cost inputs and discount rules at product level.

Promotion owner

Review profitability above 20% before approval. Check lower discounts on Tables too.

Validation and limits

Reconcile CSV and database precision.

Confirm dates, currency and profit definition.

Review product mix and discount eligibility.

Track margin and loss-making lines by region.

Observational data cannot establish causality.

Only 3 East Tables lines have no discount.

Source records extend to December 2026.

**Furniture is the review priority.
East Tables is the starting point.**

2.61%

Furniture profit margin

-11,086.15

East Tables profit

The next decision

Audit pricing and costs.
Review discount rules before
committing the next
promotion.

Wilshen Aristo Budianto
Questions & discussion